GEOGRAPHICAL INFORMATION SYSTEM

UCS 6109

Unit 1 "Introduction"

Notes by- Miss Farheen Siddiqui

Definition - Geographic Information System

A Geographic Information System (GIS) is a **computer-based framework for capturing, storing, managing, analyzing, and visualizing geographically referenced data**. By integrating hardware, software, and spatial data, it maps relationships and patterns to support decision-making across various fields like urban planning, science, and disaster management.

Key Aspects of GIS

- **Core Function:** It connects data to a map, showing where things are and what condition they are in.
- **Data Types:** Uses Vector data (points, lines, polygons) and Raster data (grid cells/pixels) to represent features.
- **Capabilities:** Allows users to create, manipulate, analyze, and display spatial data, revealing trends and relationships that are otherwise hidden.
- **Components:** Consists of software, hardware, data, and personnel.

Origin of GIS

GIS originated in the **early 1960s, pioneered by** Roger Tomlinson, who developed the first true operational system, the Canada Geographic Information System (CGIS) **in 1963**, to manage land inventory data. Rooted in quantitative geography and early computer mapping, it evolved from manual overlay techniques to digital spatial analysis.

Key Origins and Milestones

- **Conceptual Roots (1854):** Often traced back to Dr. John Snow's cholera map in London, which linked epidemiological data to location, establishing early spatial analysis.
- **The "Father of GIS" (1960s):** Roger Tomlinson initiated the field by developing CGIS for the Canadian government to analyze natural resources.

- **Technological Shift (1960s-1970s):** The rise of computing power allowed maps to move from manual paper overlays to digital storage, manipulation, and analysis.
- **Academic & Tool Development (Mid-1960s):** Early applications like SYMAP and GRID laid the foundation for raster and vector data approaches.
- **Commercialization (1980s):** The release of commercial software, notably by Esri in 1981 (ARC/INFO), helped spread GIS technology widely.

Core Drivers of Origin

1. **Quantitative Revolution:** A movement in the 1960s favoring computer programs for complex map analysis over manual methods.
2. **Need for Land Management:** Specifically, the need to process massive amounts of map data for the Canada Land Inventory.
3. **Remote Sensing Growth:** Expansion of aerial surveys and satellite data fueled the demand for systems to manage this information.

Importance of GIS

Geographic Information Systems (GIS) are vital for **collecting, managing, analyzing, and visualizing spatial data to reveal patterns, relationships, and trends**. By mapping complex info, it enables data-driven decision-making, improves operational efficiency, enhances disaster management, supports urban planning, and aids environmental monitoring.

1. Key Benefits and Importance

- **Improved Decision-Making:** Allows leaders to visualize complex data through maps, leading to more precise, informed choices.
- **Spatial Analysis & Pattern Recognition:** Helps identify relationships and trends that are not obvious in traditional data analysis.
- **Operational Efficiency:** Streamlines workflows, reduces costs, and improves record-keeping.
- **Data Integration:** Combines data from multiple sources (satellite, surveys, etc.) into one platform.

2. Core Applications & Uses

- Disaster Management: Crucial for planning, responding to, and recovering from events like floods, fires, and earthquakes by identifying high-risk areas and evacuation routes.

- Environmental Monitoring: Tracks deforestation, water pollution, soil health, and wildlife habitats.
- Urban Planning & Engineering: Assists in designing cities, managing infrastructure, planning transportation networks, and analyzing land use.
- Natural Resource Management: Manages water resources, minerals, and land cover.
- Geological Studies: Used to map fault lines, analyze seismotectonics, and assess earthquake risks.

3. Impact on Society

GIS technology is fundamental for solving complex geographic challenges, including climate change, sustainability, and social inequity. It bridges the gap between raw data and actionable intelligence.

Implementing GIS

Implementing a Geological Information System (GIS) is essential to **efficiently collect, store, analyze, and visualize complex spatial geological data, enabling better decision-making, faster analysis, and improved resource management**. It enhances geohazard assessment, mineral exploration, environmental monitoring, and infrastructure planning by identifying patterns, reducing costs, and boosting accuracy.

Key Reasons to Implement GIS in Geology

- **Enhanced Data Visualization & Mapping:** Transforms raw data, satellite imagery, and field surveys into interactive, layered maps for clearer understanding.
- **Improved Spatial Analysis:** Allows geologists to model complex relationships, identify trends, and analyze geological structures in 2D and 3D.
- **Geohazard Assessment & Management:** Helps identify and map risks from landslides, earthquakes, and volcanic activity, assisting in mitigation efforts.
- **Resource Exploration & Management:** Critical for identifying mineral deposits, managing natural resources, and analyzing soil conditions.
- **Better Decision-Making:** Provides accurate, location-based intelligence for construction, land use, and urban planning.
- **Increased Efficiency & Cost Reduction:** Speeds up data interpretation, reduces manual effort, and optimizes project timelines.

- **Data Integration:** Combines diverse datasets—geophysical, geochemical, and topographical—into one central platform.

Application of GIS or GIS in Different Disciplines

GIS (Geographic Information Systems) maps, analyzes, and models spatial data to drive decision-making across diverse sectors. Key applications include urban planning (zoning, infrastructure), environmental management (deforestation, pollution tracking), disaster management (flood modeling, risk assessment), precision agriculture, navigation, logistics, and telecommunications.

Key Applications of GIS:

- Urban Planning and Smart Cities**:** GIS helps city planners optimize land use, design infrastructure, and analyze demographics to manage urban growth.
- Environmental Management**:** It monitors natural resources, tracks pollution, and models climate change impacts.
- Disaster Management**:** GIS provides critical data for risk assessment, emergency response, and recovery, such as flood mapping.
- Agriculture**:** Used in precision farming to analyze soil, plan irrigation, and boost crop yields.
- Logistics and Navigation**:** Enables real-time tracking, route optimization for transportation, and delivery services.
- Telecommunications**:** Assists in network planning, design, and maintenance.
- Defense and Intelligence**:** Used for surveillance, terrain analysis, and strategic planning.
- Public Health**:** Tracks disease outbreaks and analyzes health data geographically.
- Natural Resource Management**:** Maps and manages forests, water resources, and mineral exploration.
- Banking and Business**:** Helps in site selection for new branches, market analysis, and risk management.

These applications rely on integrating spatial data to solve complex, location-based problems across both public and private sectors.

Map Layout

A map layout in GIS is the final, printable composition of spatial data, combining maps, legends, titles, scale bars, and north arrows onto a virtual page. Created in applications like ArcGIS Pro or QGIS, it involves setting page size, orientation (portrait/landscape), and organizing map elements for clear, professional visualization.

Key Components of a Map Layout

- **Map Frame:** The main viewport containing the map data, which can be panned and zoomed independently of the page.
- **Map Title:** Descriptive text explaining the content.
- **Legend:** Explains symbols, colors, and layers used.
- **North Arrow:** Indicates map orientation.
- **Scale Bar:** Shows distance relationships.
- **Metadata/Credits:** Information about data sources, projection, and authorship.

Workflow in GIS Software

1. **Create Layout:** Open a new layout in the software (e.g., *Insert > New Layout* in ArcGIS Pro) and set page size and orientation.
2. **Add Map Frame:** Insert the map frame(s) to the page and adjust the scale and extent to fit the map.
3. **Add Elements:** Add required elements (legend, north arrow, scale bar, text) via the "Insert" menu.
4. **Formatting:** Customize fonts, colors, and layout structure, using grids or guides for alignment.
5. **Export:** Export the final layout as a PDF, JPEG, PNG, or TIFF for printing or digital distribution.

Best Practices

- **Use Guides:** Use non-printing guides to align elements evenly.
- **Visual Hierarchy:** Ensure the main map is the focal point, with supporting elements arranged logically.
- **Dynamic Text:** Use dynamic text for elements like dates or map titles that change automatically.
- **Overview Map:** Use an inset map to show the location of the main map within a larger region.

Coordinate systems

Coordinate systems in GIS are frameworks for defining, representing, and measuring spatial locations on Earth, using either angular (latitude/longitude) or linear (planar) units. The two main types are **Geographic Coordinate Systems (GCS)**, which use a 3D spherical model (e.g., WGS84), and **Projected Coordinate Systems (PCS)**, which flatten this 3D model onto a 2D plane for mapping (e.g., UTM).

Key Aspects of GIS Coordinate Systems:

1. **Geographic Coordinate System (GCS):**
   - Uses a 3D ellipsoidal/spherical surface to define locations using degrees of latitude and longitude.
   - Locations are measured from the center of the Earth.
   - Common examples: WGS 1984 (widely used by GPS), NAD 1983.
2. **Projected Coordinate System (PCS):**
   - Defined on a 2D flat surface, using linear units like meters or feet.
   - Uses mathematical formulas to convert spherical GCS data onto a 2D plane.
   - Common examples: Universal Transverse Mercator (UTM), Albers Equal Area, State Plane.
3. **Datum:**
   - A model that defines the shape of the earth, providing a frame of reference for measurements.
   - Common datums: WGS 1984, NAD 1983.
4. **Components:**

   A coordinate system includes a coordinate axis, a unit of measurement, and a reference point (origin).
5. **Significance:**

   Proper identification of the coordinate system is crucial for integrating data from different sources and ensuring accurate measurements.

Difference between GCS and PCS:

- **GCS (3D):** Uses latitude and longitude, units in degrees, suitable for global data.
- **PCS (2D):** Uses Easting/Northing, units in meters/feet, suitable for local/regional maps to minimize distortion.

Map Scale

Map Scale is the crucial ratio comparing distance on a map to distance on the Earth, expressed as a representative fraction (e.g., 1:100,000), a verbal statement, or a graphic bar, determining the level of detail and area shown; large-scale maps show small areas with more detail, while small-scale maps cover large areas with less detail, impacting data resolution and map purpose.

Types of Scale Representation

- **Representative Fraction (RF) (Ratio):** A dimensionless ratio like 1:50,000, meaning 1 unit on the map equals 50,000 of the same units on the ground, say 1 cm on map = 50,000 cm on ground.
- **Verbal Scale:** A phrase stating the relationship, such as "1 centimeter equals 1 kilometer".
- **Graphic (Bar) Scale:** A visual ruler marked with ground distances (miles, kilometers) that remains accurate even when the map is resized.

Scale and Detail in GIS

- **Large-Scale Maps:** Have a smaller denominator (e.g., 1:10,000), showing smaller areas with high detail (buildings, roads).
- **Small-Scale Maps:** Have a larger denominator (e.g., 1:1,000,000), covering vast areas (continents) with less detail.
- **Source Scale:** The scale of the original data (aerial photo, satellite image) dictates the maximum detail available in the GIS.

Key Concepts in GIS

- **Zooming:** In GIS, zooming in on a small-scale map doesn't add accuracy; it just magnifies existing data, which can reveal pixelation if resolution is low.

- **Resolution:** Related to scale, resolution is the actual level of detail captured in the data (e.g., 1-meter resolution imagery).
- **Generalization:** When moving to smaller scales, features are simplified (e.g., a river becomes a smoother line, or small towns merge into a single point) to make the map readable.
- **Tobler's Rule:** A guideline to estimate required raster resolution: divide the map scale's denominator by 1,000 to get the detectable feature size in meters, then divide by 2 for approximate resolution.

Purpose

Choosing the right scale ensures that the map effectively serves its purpose, whether it's precise analysis (large scale) or broad overview (small scale).

Map Projection

Map projection in GIS is the mathematical transformation of the Earth's 3D curved surface onto a 2D plane (screen or paper) to represent spatial data. Because this process inherently causes distortions in area, shape, distance, or direction, selecting an appropriate projection (e.g., Cylindrical, Conic, and Azimuthal) is critical for accurate analysis and visualization, depending on the map's purpose.

Key Concepts of Map Projections

- **The Problem:** The Earth is a 3D geoid, and it is impossible to represent its surface on a 2D plane without distorting at least one of these properties: shape, area, distance, or direction.
- Georeferencing/Rectification: The process of assigning proper spatial coordinates to raster data to align it with a chosen projection system.

Types of Developable Surfaces:

- Cylindrical**:** Best for mapping the world or equatorial regions (e.g., Mercator, Transverse Mercator).
- Conic**:** Best for mid-latitude regions with a large East-West extent (e.g., Lambert Conformal Conic).
- Azimuthal/Planar**:** Best for polar regions or, when centered, for global, straight-line distance mapping.

Common Projection Types by Purpose

- Conformal**:** Preserves angles and local shapes (e.g., Mercator).
- Equal-Area (Equivalent)**:** Preserves area, but distorts shapes (e.g., Albers Equal-Area).
- Equidistant**:** Preserves distances from one or two specific points or lines.
- Compromise**:** Balances distortions to produce a visually appealing map (e.g., Robinson, Winkel Tripel).

Significance in GIS

- **Analysis:** Selecting the right projection ensures that area, distance, or direction calculations are accurate.
- **Data Integration:** Layers from different coordinate systems must be projected into a common system (reprojected) to align properly for analysis.
- **Visualization:** The choice of projection affects how users perceive the size and shape of regions.

Geometric transformation

Geometric transformation in GIS is the process of **altering the spatial location, shape, or orientation of geographic data (raster or vector) using mathematical equations and control points** to align, project, or georeference it. It involves techniques like translation, rotation, scaling, and affine transformations to move data between coordinate systems.

Key Aspects of Geometric Transformation

- Purpose: Primarily used for georeferencing (aligning images to maps), reprojection, digitizing, and correcting distortion in aerial/satellite data.
- Mechanism: Uses Ground Control Points (GCPs)—identifiable points with known coordinates—to calculate transformation equations that map old coordinates to new ones.
- Types of Changes: Can result in changes to position, direction, scale, size, or shape.

Primary Types of Transformations

- **Translation:** Moving the object/data to a new location without rotating or resizing.
- **Rotation:** Turning the data around a fixed point.
- **Scaling:** Resizing (enlarging or shrinking) the data.
- **Reflection:** Flipping the data like a mirror image.

Common GIS Transformation Techniques

- Affine Transformation**:** A versatile method that scales, rotates, skews, and translates data while preserving parallel lines. It is widely used for map-to-map or image-to-map registration.
- Georeferencing/Rectification**:** The overall process of assigning map coordinates to raster data.
- Map-to-Map Transformation**:** Converting digitized maps into a standard projected coordinate system.

Key Concepts

- Control Points (GCPs): Essential known points used to anchor the transformation.
- Resampling**:** In raster data, pixels must be reassigned to a new grid after transformation.
- Coordinate System Shift**:** Moving data from one coordinate system to another (e.g., GCS to Projected).